

BRATA-SLETTOM, Norway

WMO: 013600

Lat: 61.896N

Lon: 7.896E

Elev: 664

StdP: 93.60

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.9	-19.3	-23.1	0.5	-21.8	-20.4	0.7	-19.0	12.3	0.7	11.9	0.5	1.4	340	0.756

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.7	22.0	13.8	20.1	13.1	18.3	12.5	15.1	19.6	14.1	18.5	13.2	17.1	5.0	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.2	10.3	15.7	12.2	9.6	15.1	11.3	9.0	14.7	44.3	19.7	41.6	18.5	39.1	17.3	18.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	10.0	7.5	DB	-24.2	24.2	3.8	2.5	-26.9	26.0	-29.2	27.5	-31.3	28.9	-34.0	30.7	(4)
(5)			WB	-23.8	16.3	3.5	1.2	-26.3	17.1	-28.4	17.8	-30.4	18.4	-32.9	19.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.3	-6.7	-6.1	-3.5	1.7	6.0	9.5	12.8	12.2	7.5	2.1	-2.4	-6.0
	DBStd	8.25	6.31	6.09	4.90	3.43	3.42	2.82	2.75	2.94	3.09	3.81	5.02	6.55
	HDD10.0	3029	519	452	418	250	132	42	6	12	86	246	372	496
	HDD18.3	5856	777	685	677	500	383	265	173	191	326	503	622	754
	CDD10.0	215	0	0	0	0	7	27	92	79	9	1	0	0
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH23.3	13	0	0	0	0	0	0	12	1	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	3.3	3.3	3.3	3.9	3.6	3.7	3.7	3.3	3.4	3.4	2.8	2.8	2.9
Precipitation	PrecAvg	562	62	47	43	22	27	39	51	51	46	54	56	65
	PrecMax	804	171	124	200	53	79	101	87	91	81	133	121	154
	PrecMin	389	0	0	3	6	7	5	20	6	12	17	4	9
	PrecStd	91	41	32	39	12	16	22	20	21	19	27	35	35
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.8	5.8	6.6	11.9	19.3	22.0	25.2	23.0	16.9	12.5	9.2	6.9
		MCWB	3.5	3.4	3.7	6.5	12.0	13.4	16.2	14.6	12.1	8.9	6.3	4.7
	2%	DB	4.3	4.4	5.0	9.9	16.1	19.0	22.6	21.6	15.0	10.1	7.3	5.4
		MCWB	2.2	2.3	2.6	5.8	10.2	11.6	14.4	13.8	11.4	7.5	5.2	3.3
	5%	DB	3.0	3.1	3.9	8.3	14.2	16.8	20.6	19.9	13.5	8.4	5.0	3.9
		MCWB	1.2	1.3	1.6	4.8	9.0	10.8	13.4	13.2	10.2	6.6	3.0	2.0
	10%	DB	1.6	1.8	2.8	6.9	12.2	15.0	18.6	18.0	12.2	7.0	3.5	2.3
		MCWB	-0.1	0.2	0.7	3.9	7.9	9.9	12.8	12.9	9.4	5.5	1.8	0.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.7	3.8	4.1	7.3	13.1	14.1	17.0	16.0	13.3	9.7	6.7	5.0
		MCDB	5.5	5.6	6.1	10.7	17.4	19.9	22.0	21.5	15.2	11.9	8.7	6.7
	2%	WB	2.6	2.6	2.9	6.1	10.7	12.6	15.5	14.9	12.0	8.1	5.1	3.5
		MCDB	4.2	4.2	4.7	9.2	15.2	17.3	20.6	18.8	14.4	9.9	7.4	5.3
	5%	WB	1.4	1.5	1.9	5.0	9.6	11.6	14.5	14.0	10.8	6.9	3.2	2.3
		MCDB	3.1	3.1	3.6	7.9	13.6	15.6	19.0	18.3	13.0	8.1	4.6	3.9
	10%	WB	0.1	0.3	0.8	4.1	8.2	10.6	13.5	13.1	9.8	5.7	2.0	0.9
		MCDB	1.7	1.9	2.7	6.7	12.0	14.3	17.5	17.2	11.9	7.1	3.1	2.2
Mean Daily Temperature Range	5% DB	MDBR	4.9	5.0	5.9	6.4	7.3	7.4	7.7	7.9	6.0	3.9	4.2	4.6
		MCDBR	5.3	5.5	5.7	8.2	10.6	11.5	11.9	11.7	7.8	4.9	5.7	5.5
		MCWBR	4.6	4.7	4.4	4.9	6.0	5.6	5.2	5.1	4.6	3.5	4.6	4.5
	5% WB	MCDBR	5.4	5.3	5.5	7.4	9.5	9.6	9.7	10.0	6.6	4.9	5.6	5.3
		MCWBR	4.7	4.7	4.4	4.7	5.5	5.1	4.8	4.8	4.5	3.9	4.7	4.5
Clear-Sky Solar Irradiance	taub	0.209	0.259	0.295	0.328	0.333	0.332	0.344	0.334	0.302	0.280	0.216	0.180	
	taud	2.044	2.184	2.149	2.126	2.209	2.340	2.382	2.408	2.477	2.350	2.071	1.928	
	Ebn at Noon	499	714	810	853	882	892	870	849	818	689	474	331	
	Edh at Noon	35	65	97	123	123	111	104	93	71	55	33	22	
All-Sky Solar Radiation	RadAvg	0.21	0.77	2.05	3.80	4.85	4.93	4.62	3.55	2.19	1.00	0.30	0.13	
	RadStd	0.04	0.17	0.32	0.46	0.50	0.47	0.46	0.33	0.26	0.14	0.06	0.01	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (43 neighbors)	N/A	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page